Students Campus Laundry Booking and Management System

Requirements Specification

Functional and non-functional requirements, business rules and data requirements, with the crow's-foot entity relationship diagram (ERD) and data dictionary.

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Asemahle Sokaba (223172243) · Avuziwe Mcata (222106697)

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1. Introduction

1.1 Purpose of this document

This document states what the Campus Laundry system must do (functional requirements), how well it must do it (non-functional requirements), the rules it follows and the data it stores. It combines the problem and objectives from Phase 1 with the requirement analysis from Phase 2, and shows the database design that meets them as a crow's-foot ERD.

1.2 Background

Laundry facilities on campus are shared among many students, which often leads to overcrowding, long waiting times and poor management of machines. Students usually rely on a first-come, first-served system, which causes inconvenience and conflicts. Students also often lack knowledge about proper laundry practices such as separating clothes, choosing the correct detergent and caring for different fabric types.

1.3 Problem statement

Problem	Addressed by
Long queues and waiting times	FR-06 – FR-12 (booking and waiting list)
Poor scheduling of machines	FR-07 – FR-10, FR-23 – FR-25
No notification when laundry is done	FR-13 – FR-17
Limited knowledge of proper laundry practices	FR-18 – FR-19
No platform to give feedback on the service	FR-20 – FR-22, FR-28

1.4 Purpose and objectives of the system

The system provides a digital platform where students book laundry machines, receive notifications, read laundry tips and give feedback. Its objectives are:

- To allow students to book machines online.
- To send notifications when laundry is complete.
- To provide helpful laundry tips.
- To allow students to give feedback.
- To improve the overall management of the laundry.

1.5 Scope

In scope	Out of scope
Student registration and login; administrator accounts	Payment for laundry or detergent
Several laundry rooms with washers and dryers	Connecting to the machines themselves (students tap Start and Collected)
Booking, rescheduling, cancelling, waiting list	SMS, email or push notifications (notifications are in the app)
In-app notifications, laundry tips, feedback	A native mobile app (the web pages work on phones)
Machine and room management, booking history, usage reports	

1.6 Definitions

Term	Meaning
Slot	A fixed block of time (e.g. 60 minutes) in which one machine can be booked
Cycle	The washing or drying time of a machine, started by the student with Start cycle
No-show	A booking whose cycle was not started within 15 minutes of the slot start
Waiting list	Students waiting for a washer or dryer in a room on a day, in order of joining
Out of order	A machine with an open outage; it cannot be booked
Core table	A table that comes straight from the requirements (black heading in the ERD)
Extension table	A table added in design to meet the requirements well (grey heading in the ERD)

2. Stakeholders and users

Stakeholder	Role in the system	Main needs
Student	Main user	Book a machine without queuing; know when laundry is done; learn good laundry practice; report problems
Laundry administrator	Manages the system	Manage rooms and machines; see usage and history; handle feedback; manage accounts
Residence / campus management	Indirect	Fair, well-used facilities; reports on use and problems
Maintenance technician	Indirect	Clear reports of faulty machines (issues and outages)

2.1 Project team

Role	Responsibilities
System Analyst	Identifies user needs; analyses the system requirements
Database Designer	Designs the tables and relationships; creates the ERD
Developer	Builds the system; connects the database
Presenter	Prepares the slides; presents the project

Group members: Asemahle Sokaba (223172243) and Avuziwe Mcata (222106697).

3. Functional requirements

Priority: **Must** = from the Phase 2 requirement analysis and needed for the system to be accepted; **Should** = added in design to make the requirement work well in practice.

3.1 User management

ID	Requirement	Priority	How the system meets it
FR-01	Students register with their full name, student number, email and a password.	Must	Registration page. Student number 8–10 digits, password at least 8 characters; optional home laundry room and phone.
FR-02	Users log in securely.	Must	Log in with email or student number. Passwords stored only as bcrypt hashes; session cookie (http-only, 8 hours).
FR-03	The system supports the roles Student and Admin.	Must	Role stored per user and checked on every request; each role sees only its own menu and pages.
FR-04	Users manage their own profile.	Should	Change name, email, home room, phone and password; students can delete their own account.
FR-05	The administrator manages accounts.	Should	People page: make another admin, deactivate/reactivate, reset a password (temporary password that must be changed), delete.

3.2 Machine booking and scheduling

ID	Requirement	Priority	How the system meets it
FR-06	Students view available laundry machines in real time.	Must	Booking grid per laundry room and day (today and the next 7 days): free, taken, own and out-of-order slots.
FR-07	Students book a machine for a specific date and time slot.	Must	Fixed time slots per room (slot length set by the admin); click a free slot and confirm.
FR-08	Double bookings of the same machine are prevented.	Must	Unique key on (machine, date, slot) in the database; a student also cannot hold two machines at the same time.
FR-09	Students cancel or reschedule a booking.	Must	My laundry → Cancel or Reschedule, until the slot starts. The freed slot goes to the waiting list.
FR-10	Fair use of machines.	Should	At most 3 open bookings per student; a booking not started within 15 minutes becomes a no-show and the machine is released.
FR-11	Students record the start and end of their laundry.	Must	Start cycle (from 5 minutes before to 15 minutes after the slot starts) and Collected; the cycle time comes from the machine.
FR-12	Students join a waiting list when a room is fully booked.	Should	Waiting list per room, day and machine type; the first student in line is told when a slot frees up.

3.3 Notifications

ID	Requirement	Priority	How the system meets it
FR-13	Send a reminder when the booking time is approaching.	Must	In-app reminder 30 minutes before the slot, sent by a background scheduler.
FR-14	Notify the user when their laundry cycle is done.	Must	"Your laundry is done" when the machine's cycle time has passed after Start cycle.
FR-15	Notify the next user when a machine becomes free.	Must	After Collected, the student with the next booking on that machine today is told; freed slots go to the waiting list.
FR-16	Users see and manage their notifications.	Should	Bell with unread count on every page; notifications page with Open, Mark read, Mark all as read, Delete.

ID	Requirement	Priority	How the system meets it
FR-17	Users are told about changes made by the administrator.	Should	Notifications when a booking is cancelled (e.g. machine out of order), a slot is missed, or feedback is answered.

3.4 Laundry tips

ID	Requirement	Priority	How the system meets it
FR-18	Provide guidelines on sorting clothes, choosing detergents, washing fabric types and machine maintenance.	Must	Laundry tips page with the four categories and a filter; 8 tips included at installation.
FR-19	The administrator maintains the tips.	Should	Add, edit, delete and publish/unpublish tips (drafts are hidden from students).

3.5 Feedback system

ID	Requirement	Priority	How the system meets it
FR-20	Students rate the service or a machine's condition.	Must	Rating of 1 to 5 stars for a laundry room or a single machine.
FR-21	Students submit comments or report issues.	Must	Comment and issue forms; an issue needs a room and a description.
FR-22	The administrator views and manages feedback.	Must	Feedback page (open first): reply, set status New / Being looked at / Resolved / Closed; the student is notified. Closing needs a reply.

3.6 Management and administration

ID	Requirement	Priority	How the system meets it
FR-23	The administrator adds, edits and removes machines.	Must	Machines page. A machine that was used is retired (kept for history) instead of deleted.
FR-24	The administrator manages laundry rooms.	Should	Rooms with opening hours and slot length; hours can only change when no bookings are upcoming.
FR-25	The administrator marks machines out of order.	Should	Reason recorded; the machine can't be booked, its upcoming bookings are cancelled and the students notified; outage history kept.
FR-26	The administrator views booking history.	Must	Bookings page with filters (student, dates, room, status); the admin can cancel an upcoming booking.
FR-27	The administrator views usage reports.	Must	Dashboard: today's activity and, for 7/30/90 days, bookings, loads, cancellations, no-show rate, average rating, use per day, hour, machine and room.
FR-28	The administrator monitors feedback and takes action.	Must	Open issues on the dashboard; an issue about a machine can take it out of order in one step.

4. Non-functional requirements

ID	Quality	Requirement	How it is met and verified
NFR-01	Usability	The interface is simple, mobile-friendly and easy for students to use.	Plain-language pages with a menu (a tab bar on phones). Verified: browser tests at 1280 px and on a 375 px phone screen (no sideways scrolling).
NFR-02	Performance	The system loads quickly and handles many users booking at the same time.	Targets: availability grid under 300 ms for 95% of requests; 30 students opening "My laundry" at once answered within 2 s; a page in under 200 ms. Verified by the load tests.
NFR-03	Reliability	Booking data and notifications are accurate and consistent.	Database constraints (unique keys, CHECK rules, foreign keys); transactions with row locks; each notification sent at most once. Verified: 30 simultaneous bookings of one slot → exactly one succeeds.
NFR-04	Security	User data and passwords are stored securely.	bcrypt password hashes; http-only session cookie; parameterised SQL; role and account status re-checked on every request; temporary passwords must be changed.
NFR-05	Compatibility	Works in web browsers and on mobile devices.	Standard HTML/CSS/JavaScript; works in current Edge, Chrome and Firefox and on phones on the same network.
NFR-06	Maintainability	The database can be changed without losing data.	Versioned migrations (db/migrations) recorded in schema_migrations; a full creation script (db/schema.sql) checked against the live database.
NFR-07	Availability of data	History is kept when people leave or machines are removed.	Deleting an account keeps its bookings and feedback ("former student"); used machines are retired, not deleted.

5. Business rules

ID	Rule
BR-01	A booking is for one machine, one day and one slot of the room's fixed slot length (30–240 minutes).
BR-02	Bookings can be made for today and up to 7 days ahead, only inside the room's opening hours and never in the past.
BR-03	A student has at most 3 open bookings and cannot book two machines for the same time.
BR-04	A machine's cycle time (10–240 minutes) must fit inside one slot.
BR-05	Start cycle is possible from 5 minutes before the slot until 15 minutes after it starts; after that the booking is a no-show.
BR-06	Booking status flow: Booked → Washing/drying → Done → Collected, or Cancelled / No-show.
BR-07	A machine has at most one open outage; while it is out of order it cannot be booked.
BR-08	Feedback of kind "rating" has 1–5 stars; comments and issues have a description of at least 5 characters.
BR-09	Each booking gets each kind of notification at most once.
BR-10	Only students have a student number, and every student has one.

5.1 Assumptions and constraints

- The system runs on one computer in the residence or laundry office (Node.js and MySQL 8.4); students use it in a web browser on the same network.
- Students record the start and the collection of their laundry in the app; the system does not read the machines.
- Each laundry room has fixed opening hours and one slot length for all its machines.
- Notifications are shown in the app; they are not sent by SMS or email.

6. Data requirements and ERD

The database has 9 tables. Six come straight from the requirements (users, machines, bookings, notifications, laundry_tips, feedback). Three were added in design: **laundry_rooms** (the campus has several laundry rooms with their own hours), **waitlist_entries** (the waiting list that lets the system "notify the next user when a machine becomes free") and **machine_outages** (machines out of order, with their history).

6.1 Crow's-foot notation

Symbol at the end of a line	Meaning
Two bars (||)	Exactly one
Bar and circle (|○)	Zero or one (the link is optional)
Crow's foot and circle (○<)	Zero or many

Each line is read from the parent (top) to the child (bottom) with its verb, e.g. "a laundry room **contains** zero or many machines; a machine is in exactly one laundry room".

6.2 Conceptual ERD

The entities and how they are related.

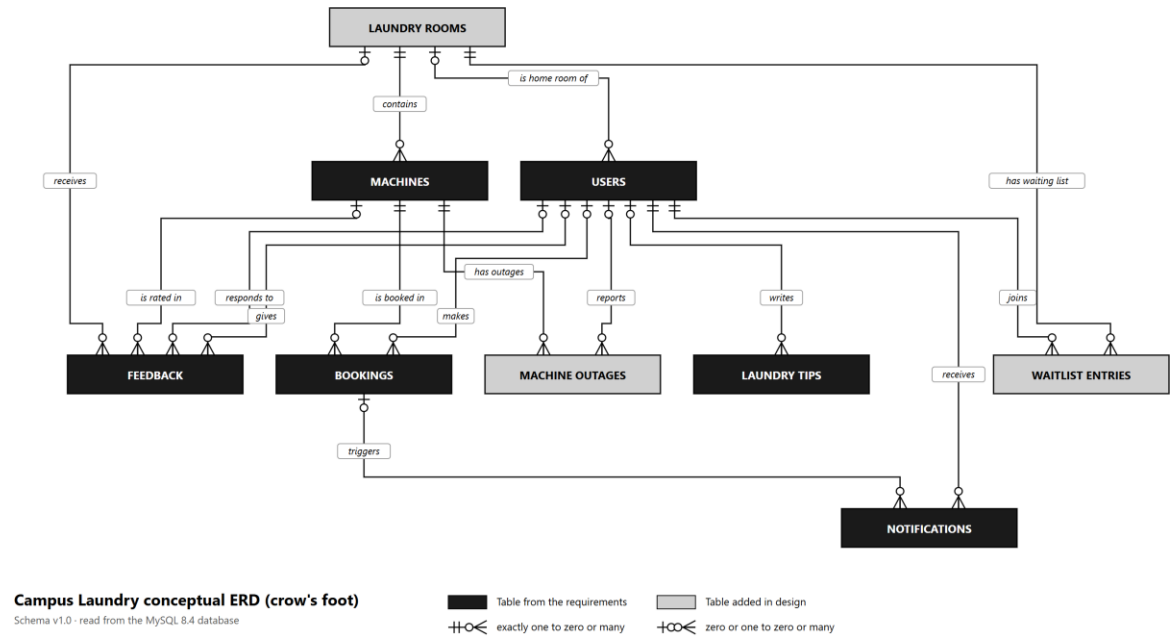


Figure 1: Conceptual ERD (crow's foot)

6.3 Physical ERD

Every table with its columns, data types and keys, as built in MySQL 8.4. The diagram is a vector image: zoom in to read it in detail.

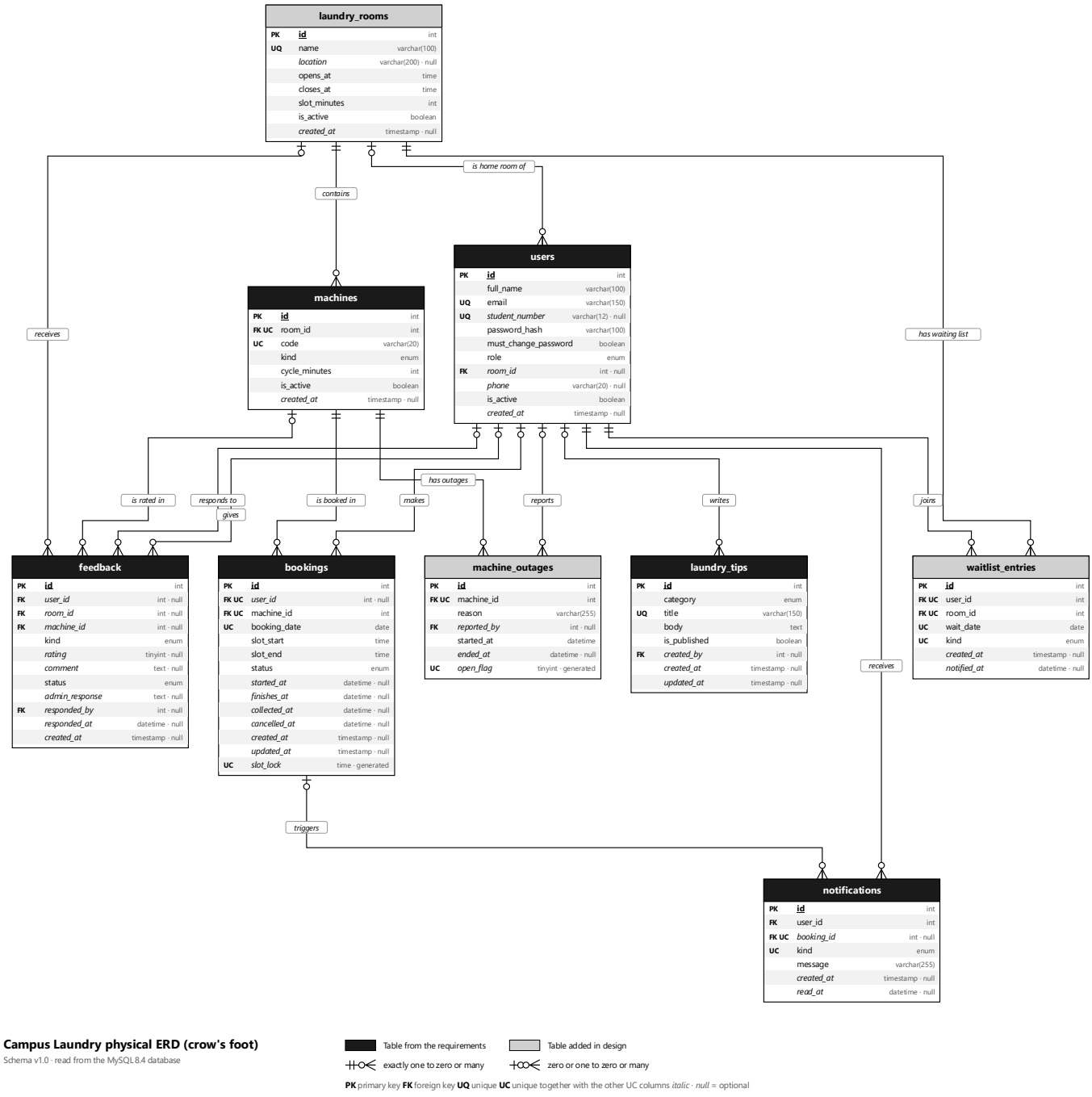


Figure 2: Physical ERD (crow's foot) with columns and keys

6.4 Relationships

Parent	Verb	Child	Cardinality	Foreign key	When the parent is deleted
machines	is booked in	bookings	1 to 0..many	bookings.machine_id	Not allowed while children exist
users	makes	bookings	0..1 to 0..many	bookings.user_id	Children are kept; the link becomes empty
machines	is rated in	feedback	0..1 to 0..many	feedback.machine_id	Children are kept; the link becomes empty
users	responds to	feedback	0..1 to 0..many	feedback.responded_by	Children are kept; the link becomes empty
laundry_rooms	receives	feedback	0..1 to 0..many	feedback.room_id	Children are kept; the link becomes empty
users	gives	feedback	0..1 to 0..many	feedback.user_id	Children are kept; the link becomes empty
users	writes	laundry_tips	0..1 to 0..many	laundry_tips.created_by	Children are kept; the link becomes empty
machines	has outages	machine_outages	1 to 0..many	machine_outages.machine_id	Children are deleted too
users	reports	machine_outages	0..1 to 0..many	machine_outages.reported_by	Children are kept; the link becomes empty
laundry_rooms	contains	machines	1 to 0..many	machines.room_id	Not allowed while children exist
bookings	triggers	notifications	0..1 to 0..many	notifications.booking_id	Children are deleted too
users	receives	notifications	1 to 0..many	notifications.user_id	Children are deleted too
laundry_rooms	is home room of	users	0..1 to 0..many	users.room_id	Children are kept; the link becomes empty
laundry_rooms	has waiting list	waitlist_entries	1 to 0..many	waitlist_entries.room_id	Children are deleted too
users	joins	waitlist_entries	1 to 0..many	waitlist_entries.user_id	Children are deleted too

6.5 Unique keys

Table	Unique key	Columns	Meaning
bookings	uq_bookings_machine_slot	machine_id, booking_date, slot_lock	A machine can't be booked twice for the same slot (ignores cancelled and missed bookings)
bookings	uq_bookings_user_slot	user_id, booking_date, slot_lock	A student can't hold two machines at the same time
laundry_rooms	uq_laundry_rooms_name	name	Room names are unique
laundry_tips	uq_laundry_tips_title	title	Tip titles are unique
machine_outages	uq_machine_outages_machine_open	machine_id, open_flag	At most one open outage per machine
machines	uq_machines_room_code	room_id, code	Machine codes are unique within a room
notifications	uq_notifications_booking_kind	booking_id, kind	Each kind of notification is sent once per booking
users	uq_users_email	email	One account per email
users	uq_users_student_number	student_number	One account per student number

Table	Unique key	Columns	Meaning
waitlist_entries	uq_waitlist_entries_user_room_date_kind	user_id, room_id, wait_date, kind	A student joins a waiting list once

6.6 Validation rules (CHECK constraints)

These rules are enforced by MySQL itself, so invalid data is refused even if it does not come through the web pages.

Table	Rule	Meaning
bookings	chk_bookings_cancelled	A cancellation time exists exactly when the status is cancelled
bookings	chk_bookings_collected	A collection time exists exactly when the status is collected
bookings	chk_bookings_finishes	Start and finish times come together, and the finish is after the start
bookings	chk_bookings_started	A start time exists exactly when the status is in progress, done or collected
bookings	chk_bookings_times	Slot end is after slot start
feedback	chk_feedback_comment	Comments and issues have a description of at least 5 characters
feedback	chk_feedback_rating	Rating is 1–5
feedback	chk_feedback_rating_kind	A rating exists exactly when the kind is "rating"
feedback	chk_feedback_response	A reply and its time come together
laundry_rooms	chk_laundry_rooms_hours	Closing time is after opening time
laundry_rooms	chk_laundry_rooms_name	Room name has at least 2 characters
laundry_rooms	chk_laundry_rooms_slot	Slot length is 30–240 minutes
laundry_tips	chk_laundry_tips_body	Tip text has at least 20 characters
laundry_tips	chk_laundry_tips_title	Tip title has at least 5 characters
machine_outages	chk_machine_outages_period	An outage ends after it starts
machine_outages	chk_machine_outages_reason	Reason has at least 5 characters
machines	chk_machines_code	Machine code is not empty
machines	chk_machines_cycle	Cycle time is 10–240 minutes
notifications	chk_notifications_message	Message has at least 5 characters
users	chk_users_email	Email looks like name@domain.tld
users	chk_users_full_name	Full name has at least 2 characters
users	chk_users_phone	Phone has 7–20 digits, spaces, +, - or brackets
users	chk_users_student_has_number	Every student has a student number
users	chk_users_student_number	Student number is 8–10 digits

7. Data dictionary

PK = primary key, FK = foreign key (→ the table it refers to), UQ = part of a unique key. "Null: Yes" means the value is optional.

7.1 users

Everyone who can log in: students and administrators. (Source: requirements.)

Column	Type	Null	Key	Description
id	int	No	PK	Unique number of the user
full_name	varchar(100)	No		Full name (at least 2 characters)
email	varchar(150)	No	UQ	Email address, used to log in
student_number	varchar(12)	Yes	UQ	Student number, 8–10 digits (students only); also used to log in
password_hash	varchar(100)	No		bcrypt hash of the password (the password itself is never stored)
must_change_password	boolean	No		True after a reset: the user must choose a new password first
role	enum	No		student or admin
room_id	int	Yes	FK → laundry_rooms	Home laundry room, opened first when booking
phone	varchar(20)	Yes		Phone number (optional)
is_active	boolean	No		False when the administrator deactivated the account
created_at	timestamp	Yes		When the account was created

7.2 laundry_rooms

A laundry room with its opening hours and slot length. (Source: design: several laundry rooms.)

Column	Type	Null	Key	Description
id	int	No	PK	Unique number of the room
name	varchar(100)	No	UQ	Name, e.g. "Residence A Laundry"
location	varchar(200)	Yes		Where it is (optional)
opens_at	time	No		Opening time
closes_at	time	No		Closing time (after opening time)
slot_minutes	int	No		Length of one booking slot, 30–240 minutes
is_active	boolean	No		False when the room is closed for bookings
created_at	timestamp	Yes		When the room was added

7.3 machines

A washer or dryer in a laundry room. (Source: requirements.)

Column	Type	Null	Key	Description
id	int	No	PK	Unique number of the machine

Column	Type	Null	Key	Description
room_id	int	No	FK → laundry_rooms, UQ	Room the machine is in
code	varchar(20)	No	UQ	Short code written on the machine, e.g. W1 (unique in its room)
kind	enum	No		washer or dryer
cycle_minutes	int	No		Length of one cycle, 10–240 minutes
is_active	boolean	No		False when the machine is retired
created_at	timestamp	Yes		When the machine was added

7.4 bookings

A student's booking of one machine for one slot, and what happened to it. (Source: requirements.)

Column	Type	Null	Key	Description
id	int	No	PK	Unique number of the booking
user_id	int	Yes	FK → users, UQ	Student who booked (empty if the account was deleted)
machine_id	int	No	FK → machines, UQ	Machine booked
booking_date	date	No	UQ	Day of the booking
slot_start	time	No		Start of the slot
slot_end	time	No		End of the slot (after the start)
status	enum	No		booked, in_progress, done, collected, cancelled or no_show
started_at	datetime	Yes		When Start cycle was tapped
finishes_at	datetime	Yes		When the cycle finishes (start + cycle time)
collected_at	datetime	Yes		When Collected was tapped
cancelled_at	datetime	Yes		When the booking was cancelled
created_at	timestamp	Yes		When the booking was made
updated_at	timestamp	Yes		When the booking last changed
slot_lock	time	Yes	UQ	Generated: the slot start for a live booking, empty when cancelled or missed (makes double bookings impossible)

7.5 notifications

A message to one user (reminder, cycle done, machine free …). (Source: requirements.)

Column	Type	Null	Key	Description
id	int	No	PK	Unique number of the notification
user_id	int	No	FK → users	Who receives it
booking_id	int	Yes	FK → bookings, UQ	Booking it is about (optional)
kind	enum	No	UQ	reminder, cycle_done, machine_free, booking_cancelled, no_show or feedback_reply
message	varchar(255)	No		Text shown to the user
created_at	timestamp	Yes		When it was sent

Column	Type	Null	Key	Description
read_at	datetime	Yes		When it was read (empty = unread)

7.6 waitlist_entries

A student waiting for a washer or dryer in a room on a day. (Source: design: waiting list.)

Column	Type	Null	Key	Description
id	int	No	PK	Unique number of the entry
user_id	int	No	FK → users, UQ	Student waiting
room_id	int	No	FK → laundry_rooms, UQ	Room
wait_date	date	No	UQ	Day
kind	enum	No	UQ	washer or dryer
created_at	timestamp	Yes		When the student joined (decides the order)
notified_at	datetime	Yes		When the student was told a machine is free

7.7 machine_outages

A period during which a machine is out of order. (Source: design: out-of-order machines.)

Column	Type	Null	Key	Description
id	int	No	PK	Unique number of the outage
machine_id	int	No	FK → machines, UQ	Machine that is out of order
reason	varchar(255)	No		Why (at least 5 characters)
reported_by	int	Yes	FK → users	Administrator who recorded it
started_at	datetime	No		When it went out of order
ended_at	datetime	Yes		When it was back in service (empty = still out of order)
open_flag	tinyint	Yes	UQ	Generated: 1 while the outage is open (allows only one open outage per machine)

7.8 laundry_tips

A laundry tip in one of the four categories. (Source: requirements.)

Column	Type	Null	Key	Description
id	int	No	PK	Unique number of the tip
category	enum	No		sorting, detergents, fabrics or maintenance
title	varchar(150)	No	UQ	Title (unique)
body	text	No		The tip (at least 20 characters)
is_published	boolean	No		False for a draft that students don't see
created_by	int	Yes	FK → users	Administrator who wrote it
created_at	timestamp	Yes		When it was written
updated_at	timestamp	Yes		When it was last changed

7.9 feedback

A rating, comment or issue from a student, and the administrator's reply. (Source: requirements.)

Column	Type	Null	Key	Description
id	int	No	PK	Unique number of the feedback
user_id	int	Yes	FK → users	Student who sent it
room_id	int	Yes	FK → laundry_rooms	Laundry room it is about
machine_id	int	Yes	FK → machines	Machine it is about (optional)
kind	enum	No		rating, comment or issue
rating	tinyint	Yes		1–5 stars (ratings only)
comment	text	Yes		Comment or description of the issue
status	enum	No		new, in_progress, resolved or dismissed
admin_response	text	Yes		The administrator's reply
responded_by	int	Yes	FK → users	Administrator who replied
responded_at	datetime	Yes		When the reply was sent
created_at	timestamp	Yes		When the feedback was sent

8. Requirements traceability

Where each group of requirements is stored and which pages provide it.

Requirements	Tables	Pages
FR-01 – FR-05	users	Register / Log in, Profile, People
FR-06 – FR-11	laundry_rooms, machines, bookings	Book a machine, My laundry
FR-12	waitlist_entries	Book a machine, My laundry
FR-13 – FR-17	notifications	Bell, Notifications
FR-18 – FR-19	laundry_tips	Laundry tips
FR-20 – FR-22	feedback	Feedback
FR-23 – FR-25	machines, laundry_rooms, machine_outages	Machines & rooms
FR-26 – FR-28	bookings, feedback (read)	Dashboard, Bookings, Feedback

> All requirements are verified by automated tests: database constraint tests, end-to-end API tests, browser tests and concurrency/performance tests (see the Test Report).